

A 40 year old proof of a 65 year old theorem

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References and teaser

1. Minoru Nakaoka, *Decomposition theorems for homology of symmetric groups*, Ann. of Math. **71** (1960) 16–42.
2. Albrecht Dold, *Decomposition theorems for $S(n)$ -complexes*, Ann. of Math. **75** (1962) 8–16.
3. André Joyal, *Calcul integral combinatoire et homologie des groupes symetriques*, C.R. Acad. Sci. Canada **VII(6)**(Dec. 1985) 337–342.
 - ▶ The Theorem comes from [1] with a different approach in [2]. The Proof today comes from the [3].
 - ▶ While the inclusions $\mathfrak{S}_n \hookrightarrow \mathfrak{S}_{n+1}$ of symmetric groups are not retractions, they do induce retractions on homology.
Questo è il Teorema!
 - ▶ The Proof in question uses the calculus of virtual Joyal species in a clever way. Suffice it to say, at this point, $\#\mathfrak{S}_n = n!$ and the shift $n \mapsto n + 1$ features in taking the derivative of a Taylor series.

Acknowledgement

I was visiting SUNY Buffalo in 1985. On 21 June 1985, Bill Lawvere gave a seminar on Joyal's work. Presumably he had a preprint of the *Comptes Rendues* article which appeared later in the year. This talk is based on André's article and Bill's seminar.

A formula for antiderivative

- Suppose we have a sequence of functions $f_0, f_1, f_2, f_3, \dots$ such that f_{n+1} is the derivative of f_n for all $n \geq 0$, and $g_0, g_1, g_2, g_3, \dots$ such that g_n is the derivative of g_{n+1} for all $n \geq 0$. Then the rule for differentiating a product gives:

$$f_0(x) g_0(x) = \frac{d}{dx} \sum_{k \geq 0} (-1)^k f_k(x) g_{k+1}(x) .$$

- In particular, we obtain a Taylor series for an antiderivative of $f(x)$ in terms of its derivatives by taking $f_0(x) = f(x)$ and $g_n(x) = \frac{x^n}{n!}$:

$$\begin{aligned} \int f(x) dx &= \sum_{k \geq 0} (-1)^k f^{(k)}(x) \frac{x^{k+1}}{(k+1)!} \\ &= f(x)x - f'(x)\frac{x^2}{2!} + f''(x)\frac{x^3}{3!} - \dots \end{aligned}$$

Joyal species for objective combinatorics

- ▶ $\mathfrak{S}$ is the symmetric groupoid: its objects are natural numbers, all morphisms are endomorphisms, while $\mathfrak{S}(n, n)$ is the *symmetric group* $\mathfrak{S}_n$ of permutations on the n symbols $1, 2, \dots, n$.
- ▶ A *species (of structure)* is a functor $F : \mathfrak{S} \rightarrow \mathbf{Set}$. The value at n is written $F[n]$ and is thought of as the set of F -structures on the set $Jn = \{1, 2, \dots, n\}$.
- ▶ There is a faithful inclusion functor $J : \mathfrak{S} \rightarrow \mathbf{Set}$ taking n to $\{1, 2, \dots, n\}$. The left Kan extension of $F : \mathfrak{S} \rightarrow \mathbf{Set}$ along J is denoted by $F : \mathbf{Set} \rightarrow \mathbf{Set}$ and its value at the set X is given by the objective Taylor series

$$F(X) = \int^{n \in \mathfrak{S}} \mathbf{Set}(Jn, X) \times F[n] \cong \sum_{n \geq 0} X^n \times_{\mathfrak{S}_n} F[n] .$$

Two easy facts about coends and conical colimits

- ▶ If $T : \mathcal{A} \rightarrow \mathcal{X}$ is the terminal functor $TA = 1$ then a coend of the form $\int^A SA \times TA$ is the colimit of $S : \mathcal{A}^{\text{op}} \rightarrow \mathcal{X}$.
- ▶ If $\mathcal{A}$ is a groupoid and $(-)^* : \mathcal{A} \rightarrow \mathcal{A}^{\text{op}}$ denotes the isomorphism $A \mapsto A, f \mapsto f^{-1}$ then $\int^A K(A, A)$ is the colimit of the composite

$$\mathcal{A} \xrightarrow{\Delta} \mathcal{A} \times \mathcal{A} \xrightarrow{(-)^* \times 1} \mathcal{A}^{\text{op}} \times \mathcal{A} \xrightarrow{K} \mathcal{X}.$$

That is, $\int^A K(A, A) \cong \text{colim}_A K(A^*, A)$.

- ▶ As a particular case of the last point,

$$F(X) \cong \sum_{n \geq 0} (X^{n*} \times F[n]) / \mathfrak{S}_n.$$

- ▶ If F is the terminal functor in this last then the first point gives

$$\exp(X) \cong \text{colim}_{n \in \mathfrak{S}} \text{Set}(Jn, X) \cong \sum_{n \geq 0} X^{n*} / \mathfrak{S}_n$$

for the *exponential species*.

Differentiation of species

- ▶ The derivative F' of a species F is its composite with the endofunctor of $\mathfrak{S}$ taking n to $n + 1$ and using the obvious inclusion $\mathfrak{S}_n \rightarrow \mathfrak{S}_{n+1}$ on morphisms; so $F'[n] = F[n + 1]$.
- ▶ This is justified by the objective form

$$F'(X) = \sum_{n \geq 0} F[n + 1] \times_{\mathfrak{S}_n} X^n$$

of the usual (abstract) formula for the derivative of a Taylor series.

- ▶ We have the familiar looking formulas

$$\frac{d}{dX}(F(X) + G(X)) \cong F'(X) + G'(X) ,$$

$$\frac{d}{dX}(F(X)G(X)) \cong F'(X)G(X) + F(X)G'(X) .$$

Virtual species

- ▶ Those formulas for derivative are enough to prove a species version of our formula for antiderivative in terms of derivatives **except for one problem: minus signs**. We need to move to *virtual species*.
- ▶ The main fact that makes virtual species work is already apparent for *virtual sets*: that is, pairs (P, M) of sets, with M finite, modulo the relation $(P, M) \sim (Q, N)$ if $P + N \cong Q + M$. Now, from a bijection $\phi : X + U \cong Y + U$ with U finite, we can construct a **trace bijection** $\psi : X \cong Y$ where $\psi(x) = \phi^n(x)$ where $n > 0$ gives the time $\phi^n(x) \notin U$. So we have an equivalence relation, because $(P, M) \sim (Q, N) \sim (R, L)$ implies $P + N \cong Q + M$ and $R + M \cong Q + N$, so $P + L + N \cong R + M + N$, which then gives $(P, M) \sim (R, L)$. We write $P - M$ for the equivalence class of (P, M) .
- ▶ With these ingredients, we can make sense of

$$F(X) \cong \frac{d}{dX} \sum_{k \geq 0} (-1)^k (X^{k+1} / \mathfrak{S}_{k+1}) \times F^{(k)}(X) .$$

Homological functors

Let $\mathbf{gpd}$ denote the category of **finite groupoids** and let $\mathcal{A}$ be an additive category. Here a *homological functor* $H : \mathbf{gpd} \rightarrow \mathcal{A}$ is to be equipped with a *transfer functor* $\Sigma : \mathbf{gpd}_{\text{dof}}^{\text{op}} \rightarrow \mathcal{A}$ which agrees with H on objects and is defined on the subcategory $\mathbf{gpd}_{\text{dof}}$ of $\mathbf{gpd}$ with morphisms the discrete opfibrations. The following properties are required.

- ▶ If there is a natural transformation from f to g then $Hf = Hg$.
- ▶ If the square

$$\begin{array}{ccc} P & \xrightarrow{u} & B \\ p \downarrow & & \downarrow q \\ A & \xrightarrow{v} & C \end{array}$$

is a pullback with q a discrete opfibration then $\Sigma_q \circ Hv = Hu \circ \Sigma_p$.

- ▶ H preserves finite coproducts: $H(A + B) \cong HA \oplus HB$, $H\emptyset \cong 0$.

Examples

- (i) Let $\mathcal{C}$ be a category with finite coproducts. For $A \in \mathbf{gpd}$, form the functor category $\mathbf{H}A = [A, \mathcal{C}]$ to obtain a 2-functor $\mathbf{H} : \mathbf{gpd} \rightarrow \mathbf{Cat}_+^{\text{op}}$. For a discrete opfibration $p : X \rightarrow A$, there is a left adjoint functor $\Sigma_p \dashv \mathbf{H}p$. Composing with the Grothendieck abstraction 2-functor $K_0 : \mathbf{Cat}_+ \rightarrow \mathbf{Abgp}$ into abelian groups, we obtain a homological functor with values in the additive category $\mathbf{Abgp}^{\text{op}}$.
- (ii) For $p \in \mathbf{gpd}_{\text{dof}}$ and u in $\mathbf{Gpd}$ with C an abelian group, define Σ_p by $\Sigma_p(u)a = \sum_{p(x)=a} u(\alpha_*x)$ for $a \in A$ and put $HA = \pi_0[A, C]$ with pointwise addition.

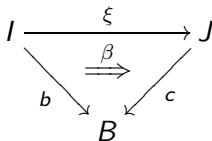
$$\begin{array}{ccc}
 X & \xrightarrow{p} & A \\
 & \searrow u & \swarrow \Sigma_p(u) \\
 & C &
 \end{array}$$

Strategy

- ▶ The idea is to construct the universal homological functor and prove the result for it. Then it will hold for all homological functors since all functors preserve retracts.
- ▶ We will look at an objective homological pseudofunctor from which the universal functor can be abstracted using the Grothendieck construction K_0 of an abelian group from a category with coproducts.
- ▶ We will work with the 2-category $\mathbf{cat}$ of finite categories as far as possible rather than finite groupoids.

Finite coproduct completion

- ▶ The finite coproduct completion $\mathbf{fam} B$ of a finite category B is the closure of the image of the Yoneda embedding $\mathcal{Y}_B : B \rightarrow [B^{\text{op}}, \mathbf{set}]$ under finite coproducts.
- ▶ There is a bijection between natural transformations $\sum_{i \in I} B(-, b_i) \rightarrow \sum_{j \in J} B(-, c_j)$ between finite coproducts of representables and diagrams

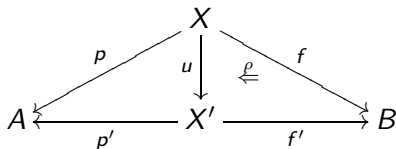


where $\xi : I \rightarrow J$ is a function between finite sets.

- ▶ We can identify $\mathbf{fam} B$ with the category whose objects are finite families $I \xrightarrow{b} B$ of objects of B and whose morphisms are $(\xi, \beta) : b \rightarrow c$ as in the diagram.

Partial functors

- ▶ A *partial functor* between finite categories A and B is a span $A \xleftarrow{p} X \xrightarrow{f} B$ in $\mathbf{cat}$ where p is a discrete opfibration.
- ▶ There is a bicategory $\mathbf{parf}$ whose objects are finite categories, morphisms are partial functors, composition is by pullback of spans, and 2-cells are as in the diagram.



Partial functors as modules

- ▶ Left Kan extension exhibits $\mathbf{fam}B$ as the partial functor classifier.

$$R(a) = \sum_{p(x)=a} B(-, f(x))$$

$$\begin{array}{ccc} X & \xrightarrow{p} & A \\ f \downarrow & \xRightarrow{\kappa} & \downarrow R \\ B & \longrightarrow & \mathbf{fam}B \end{array}$$

- ▶ The inclusion $\mathbf{fam}B \hookrightarrow [B^{\mathrm{op}}, \mathbf{set}]$ shows that every partial function gives a module $R : B^{\mathrm{op}} \times A \rightarrow \mathbf{set}$ where $R(a)b$ becomes $R(b, a)$.
- ▶ This gives a fully faithful pseudofunctor $\mathbf{parf} \rightarrow \mathbf{mod}$.
- ▶ Partial functors play a universal role for homological functors akin to spans for Mackey functors and polynomials for Tambara functors.

Abstraction

- ▶ The homs $\text{parf}(A, B)$ have finite coproducts preserved by composition on both sides. Put $\text{abparf}(A, B) = K_0\text{parf}(A, B)$ to obtain an additive category abparf .
- ▶ Define $H : \text{gpd} \rightarrow \text{abparf}$ by $H(A \xrightarrow{f} B) = [A \xleftarrow{1_A} A \xrightarrow{f} B]$.
- ▶ Define $\Sigma : \text{gpd}_{\text{dof}}^{\text{op}} \rightarrow \text{abparf}$ by $H(A \xrightarrow{p} B) = [B \xleftarrow{p} A \xrightarrow{1_A} A]$.

Proposition

$\text{abparf}(A, B)$ with this H and Σ is universal amongst homological functors.

A bit more explicitly, restriction along H gives an equivalence between the category of additive functors $\text{abparf} \rightarrow \mathcal{A}$ and the category of homological functors $\text{gpd} \rightarrow \mathcal{A}$.

Retraction condition

- ▶ For each $n \geq 0$, suppose $\mathfrak{S}_{n+1} \xrightarrow{r_n} \mathfrak{S}_n$ is a retraction for the inclusion $\mathfrak{S}_n \xrightarrow{i_n} \mathfrak{S}_{n+1}$ in parf .
- ▶ Define a Set-module $\mathfrak{S} \xrightarrow{R} \mathfrak{S}$ by

$$R[m, n] = \begin{cases} r_n & \text{if } n = m + 1 \\ 0 & \text{otherwise .} \end{cases}$$

- ▶ The retract condition is expressed by commutativity of the triangle

$$\begin{array}{ccc} \mathfrak{S} & \xrightarrow{-+1} & \mathfrak{S} \\ & \searrow \mathfrak{S}(-,-) & \swarrow R \\ & \mathfrak{S} & \end{array}$$

of modules. That is,

$$R[m, k + 1] = \begin{cases} \mathfrak{S}(m, k) & \text{if } m = k \\ 0 & \text{otherwise .} \end{cases}$$

Finding R

- ▶ We have a 2-variable species

$$R(X, Y) = \int^{m,n} \text{Set}(Jn, X) \times \text{Set}(Jm, Y) \times R[m^*, n] .$$

- ▶ Using the property of R from the last frame, we have

$$\begin{aligned} \frac{d}{dX} R(X, Y) &\cong \int^{m,n} \text{Set}(Jn, X) \times \text{Set}(Jm, Y) \times R[m^*, n+1] \\ &\cong \int^{m,n} \text{Set}(Jn, X) \times \text{Set}(Jm, Y) \times \mathfrak{S}(m^*, n) \\ &\cong \int^m \text{Set}(Jm^*, X) \times \text{Set}(Jm, Y) \\ &\cong \text{colim}_m \text{Set}(Jm, X) \times \text{Set}(Jm, Y) \\ &\cong \int^m \text{Set}(Jm, X \times Y) \\ &\cong \exp(X \times Y) . \end{aligned}$$

Formula for R determining the r_n

- ▶ As you might expect

$$\frac{d^k}{dX^k} \exp(X \times Y) \cong \exp(X \times Y) \times Y^k .$$

- ▶ Applying our antiderivative formula to $\exp(X \times Y)$ we obtain

$$R(X, Y) \cong \exp(X \times Y) \times \sum_{k \geq 0} (-1)^k (X^{k+1} / \mathfrak{S}_{k+1}) \times Y^k .$$

Thank you one and all!

